PROGRAM CODE:

```
clear;
clc;
xdel(winsid());
mode(0);
Mp=0.2;
tp=1;
J=1;//kg.mˆ2
B=1;//N−/rad/sec
s=%s;
pi=%pi;
Eq=(s*pi)^2-log(1/Mp)^2*(1-s^2);
disp(Eq, 'The obtained equation')
x=roots(Eq);
disp(x,'x=')
zeta=abs(x(1));
disp(zeta,'zeta=')
wd=pi/tp;
disp(wd,'damped frequency wd=')
wn=wd/sqrt(1-zeta^2);
disp(wn,'natural frequency wn=')

K=J*wn^2;
disp(K, 'K=')
Kh=(2*sqrt(K*J)*zeta-B)/K;
disp(Kh,'Kh=')
sigma=wn*zeta;
disp(sigma, 'sigma=')
```

```
_beta=atan(wd/sigma);
disp(_beta,'Beta =')
tr=(pi-_beta)/wd;
disp(tr,'Rise time tr=')
ts_2percent=4/sigma;
disp(ts_2percent, 'Settling time for 2% error=')
ts_5percent=3/sigma;
disp(ts_5percent, 'Settling time for 5% error=')
```

Scilab 2025.1.0 Console

```
 -2.5902904 +12.459895s^2
 "The obtained equation:"
  0.4559498 + 0.i
 -0.4559498 + 0.i
 "Roots x ="
  0.4559498
 "Damping ratio zeta ="
  3.1415927
 "Damped frequency wd ="
  3.5298576
 "Natural frequency wn ="
  12.459895
 "K ="
  0.1780814
 "Kh ="
  1.6094379
 "Sigma ="
  1.0973572
 "Beta ="
  0.6507004
 "Rise time tr ="
  2.4853397
 "Settling time for 2% error ="
  1.8640048
 "Settling time for 5% error ="

-->
```